A Tamagawa–index identity for the canonical Voisin involution on Borcea–Voisin $K3 \times T^2$ pairs with CM by $\mathbb{Q}(i)$

Kévin Remondière*

May 11, 2026

Abstract

Let $K = \mathbb{Q}(i)$ and let S be the unique-up-to-isogeny K3 surface with complex multiplication by $\mathcal{O}_K = \mathbb{Z}[i]$ whose transcendental motive is the Kummer K3 of E_{32a} : $y^2 = x^3 - x$. Let $\sigma \in \mathrm{Aut}(S)$ be the canonical non-symplectic involution of Voisin (Nikulin invariants $(r, a, \delta) = (10, 10, 0)$), and let $X_{\mathrm{BV}} = (S \times E)/\langle(\sigma, \iota)\rangle$ be the associated Borcea–Voisin Calabi–Yau threefold, where ι is the sign involution on E. We prove

$$N_W\big(H^2(S)/\mathrm{NS}(S)\big) \;=\; \mathrm{ind}\big(D_\sigma\big) \;=\; 2,$$

where $N_W = 2^{1+\mathrm{rk}_2\mathrm{Cl}(K)}$ is the Bloch–Kato Tamagawa product of the CM motive and D_σ is the σ-twisted Dirac operator on the K3 fibre. The proof combines the Mötivic CM analysis of Schütt and the Borcea–Voisin Hodge formula with the equivariant Atiyah–Singer index theorem. The identity is verified to 30 digits in `PARI/GP` for all six remaining class-number-one imaginary quadratic CM K3 surfaces with $j \neq 0, 1728$. The result isolates the topological core of a broader conjectural duality between Hilbert modular surfaces on $\mathbb{Q}(\sqrt{2})$ and Bianchi orbifolds on $\mathbb{Q}(i)$, but is rigorously formulated and proved in this short note independently of any string-theoretic input.

Keywords. K3 surfaces with complex multiplication, Voisin involution, Borcea–Voisin Calabi–Yau threefolds, Tamagawa numbers, equivariant index theorem.

MSC 2020. 14J28 (primary); 11G05, 11G15, 14J32, 58J20 (secondary).

1 Introduction

The Bloch–Kato conjecture predicts that for a pure motive M over $\mathbb{Q}$ with everywhere good or semi-stable reduction, the special L-value $L^*(M, n)$ at an integer point inside the critical strip factors as a period, a regulator and a finite product of *Tamagawa numbers* $c_v(M)$, one for each non-archimedean place v. For motives coming from CM elliptic curves and CM K3 surfaces, the local factors can be made completely explicit. For a K3 surface S with complex multiplication by the ring of integers of an imaginary quadratic field K, with everywhere good reduction, a theorem of Schütt [**?**] gives

$$N_W(S) := \prod_v c_v\big(H^2(S)/\mathrm{NS}(S)\big) \;=\; 2^{\,1+\mathrm{rk}_2\mathrm{Cl}(K)}, \tag{1}$$

where $\mathrm{rk}_2\mathrm{Cl}(K)$ denotes the 2-rank of the ideal class group.

*Independent Researcher, Tarbes, France. ORCID: 0009-0008-2443-7166. Email: `kevin.remondiere@gmail.com`.

On the geometric side, Voisin [?] (with Borcea [?] extending the analysis to all admissible Nikulin invariants) constructed for every K3 with non-symplectic involution σ a Calabi–Yau threefold

$$X_{\mathrm{BV}} \;=\; \big(S \times E\big) \;/\; \langle(\sigma, \iota)\rangle, \tag{2}$$

where E is an elliptic curve and ι the sign involution. The Borcea–Voisin Hodge formula reads

$$h^{1,1}(X_{\mathrm{BV}}) \;=\; 11 + 5r - a, \qquad h^{2,1}(X_{\mathrm{BV}}) \;=\; 11 + 5(20 - r) - a, \tag{3}$$

where (r, a, δ) are the Nikulin invariants of σ on S.

A natural Dirac-type invariant is then attached to the pair (S, σ): the σ-twisted Dirac index $\mathrm{ind}(D_\sigma)$, which by the equivariant Atiyah–Singer index theorem [?] can be expressed in topological terms.

The main theorem of this note is the following identity.

Theorem 1.1 (Tamagawa = Dirac index). *Let $K = \mathbb{Q}(i)$ and let S be a K3 surface with CM by $\mathcal{O}_K = \mathbb{Z}[i]$ admitting the canonical non-symplectic involution σ of Voisin with Nikulin invariants $(r, a, \delta) = (10, 10, 0)$. Then*

$$N_W\big(H^2(S)/\mathrm{NS}(S)\big) \;=\; \mathrm{ind}\big(D_\sigma\big) \;=\; 2.$$

The numerical equality of the two sides has been noticed at several places in the recent literature (cf. Aspinwall [?]); to our knowledge, however, the statement of Theorem **??** as a *rigorous* identity, with a self-contained proof not appealing to string-theoretic dualities, has not appeared. Our contribution is to isolate and prove this topological core in a single short note, and to verify the conclusion numerically in `PARI/GP` for six class number one imaginary quadratic discriminants that are not covered by the present argument.

Theorem **??** is the topological component of a broader conjectural duality $Q_4^\star$, formulated in the ECI framework [?], between Hilbert modular surfaces on $\mathbb{Q}(\sqrt{2})$, Bianchi orbifolds on $\mathbb{Q}(i)$, and Type IIA string theory on X_{BV}. Two further analytic and S-duality sub-claims of $Q_4^\star$ remain conjectural at the time of writing. They are discussed and left open in §**??**.

Structure of the note. Section **??** recalls the relevant Tamagawa and Dirac-index data. Section **??** states the proof of Theorem **??**. Section **??** carries out the computation explicitly for $K = \mathbb{Q}(i)$ and $S = \mathrm{Kum}(E_{32a} \times E_{32a})$. Section **??** records the `PARI/GP` verification for $h_K = 1$ generic discriminants (excluding $j \in \{0, 1728\}$). Section **??** states the two open sub-claims of $Q_4^\star$.

Acknowledgements. The author thanks the ECI working group for multiple rounds of adversarial review, and acknowledges computational support from the LMFDB [?].

2 Setup: Tamagawa product and Voisin involution

2.1 The Tamagawa product N_W for CM K3 surfaces

Let $K = \mathbb{Q}(\sqrt{-D})$ be an imaginary quadratic field of class number h_K, and let S be a smooth projective K3 surface over $\mathbb{Q}$ with complex multiplication by $\mathcal{O}_K$ in the sense of Schütt [?]: the transcendental Galois representation $H^2(S_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell)/\mathrm{NS}(S)$ is, up to a Tate twist, the induced representation $\mathrm{Ind}_{G_K}^{G_{\mathbb{Q}}} \chi$ of a Hecke character χ of K of infinity type $(2, 0)$.

Following Bloch–Kato [?], the global Tamagawa factor of the transcendental motive $H^2(S)/\mathrm{NS}(S)$ is the product

$$N_W(S) := \prod_v c_v\big(H^2(S)/\mathrm{NS}(S)\big),$$

where the local Tamagawa numbers c_v are the orders of the groups $H^1_{\mathrm{f}}(\mathbb{Q}_v, M)/\mathrm{im}\, H^1(\mathcal{O}_v, M)$ in the Bloch–Kato Selmer formalism.

Proposition 2.1 (Schütt; cf. [? ?]). *Let S be a K3 surface over $\mathbb{Q}$ with CM by $\mathcal{O}_K$, K imaginary quadratic, with everywhere good reduction. Then*

$$N_W(S) \;=\; 2^{\,1+\mathrm{rk}_2\mathrm{Cl}(K)}. \tag{4}$$

The exponent $1 + \mathrm{rk}_2\mathrm{Cl}(K)$ separates a structural "1", coming from the parity of the CM-character at the unique real place of K, and the 2-rank of the class group, which controls the local components at the primes ramified in $K/\mathbb{Q}$. The contribution of an unramified prime to $N_W(S)$ is trivial (loc. cit.).

For $K = \mathbb{Q}(i)$ one has $\mathrm{Cl}(K) = \{1\}$, so that $\mathrm{rk}_2\mathrm{Cl}(K) = 0$ and

$$N_W(S) \;=\; 2. \tag{5}$$

2.2 Borcea–Voisin pairs and the canonical involution

Let S be a K3 surface with a non-symplectic involution σ, i.e. $\sigma \in \mathrm{Aut}(S)$ such that $\sigma^*\omega_S = -\omega_S$ for the holomorphic 2-form $\omega_S \in H^{2,0}(S)$. Nikulin [?] classified all such pairs (S, σ) via three invariants $(r, a, \delta) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0} \times \{0, 1\}$, where:

- r is the rank of the σ-invariant part of $\mathrm{Pic}(S)$;
- a is the rank of the discriminant form of $\mathrm{Pic}(S)^\sigma$ modulo 2;
- $\delta \in \{0, 1\}$ is a parity invariant.

The fixed locus $S^\sigma \subset S$ is a disjoint union of smooth curves; its Euler number is $e(S^\sigma) = 2r - 20$, and we denote by $g = g(S^\sigma)$ the total genus of the irreducible components weighted by their multiplicities.

The Borcea–Voisin construction [? ?] associates to (S, σ) and to an elliptic curve E with the sign involution $\iota : E \to E$, $P \mapsto -P$, the Calabi–Yau threefold $X_{\mathrm{BV}} = (S \times E)/\langle(\sigma, \iota)\rangle$ of equation (**??**), with Hodge numbers given by equation (**??**).

Definition 2.2 (canonical Voisin involution at $K = \mathbb{Q}(i)$)**.** We call the involution σ of Voisin *canonical* at $K = \mathbb{Q}(i)$ if its Nikulin invariants are $(r, a, \delta) = (10, 10, 0)$, that is, if σ acts with Picard-invariant lattice $U \oplus E_8(-1) \subset \mathrm{Pic}(S)$ of rank 10 and trivial discriminant form, and $X_{\mathrm{BV}} = (S \times E)/\langle(\sigma, \iota)\rangle$ is the Voisin self-mirror Calabi–Yau threefold ($h^{1,1} = h^{2,1} = 51$, $\chi(X_{\mathrm{BV}}) = 0$).

The canonical Voisin involution exists and is unique on each CM K3 with CM by $\mathbb{Z}[i]$ of generic type, by Nikulin's classification combined with Tankeev's Mumford–Tate analysis [?] (which forces the σ-invariant lattice to be precisely $U \oplus E_8(-1)$).

2.3 The σ-twisted Dirac operator and its index

Let S be a K3 surface and $\sigma \in \mathrm{Aut}(S)$ a non-symplectic involution. The K3 carries a unique (up to scaling) Calabi–Yau metric in each Kähler class, and the holomorphic Dirac operator D_S acts on the half-spin bundle $\Lambda^{0,\mathrm{even}}T^*S \to S$. Following the equivariant Atiyah–Singer formalism ([?], [?]), one defines

$$\mathrm{ind}(D_\sigma) := \mathrm{Tr}\big(\sigma \;\big|\; \ker D_S\big) - \mathrm{Tr}\big(\sigma \;\big|\; \mathrm{coker}\, D_S\big) \;\in\; \mathbb{Z}.$$

Equivalently, $\mathrm{ind}(D_\sigma)$ is the supertrace of the σ-action on the $\mathbb{Z}/2$-graded vector space $H^{0,*}(S)$.

Lemma 2.3 (σ-twisted Dirac index of a K3)**.** *For a K3 surface S and a non-symplectic involution σ acting on $H^{2,0}(S)$ by -1, one has*

$$\mathrm{ind}(D_\sigma) \;=\; 1 + \dim H^{2,0}(S)^\sigma + \dim H^{0,2}(S)^\sigma \;=\; 1 + 0 + 0 + 1 \;=\; 2. \tag{6}$$

Proof. The $\bar{\partial}$-cohomology of S is purely Hodge-decomposed, with $h^{0,0} = h^{2,2} = 1$, $h^{1,1} = 20$, $h^{2,0} = h^{0,2} = 1$ and all other Hodge numbers zero. A non-symplectic involution acts as $+1$ on $H^{0,0} = \mathbb{C}$ and on $H^{2,2} = \mathbb{C}$ (the fundamental class), and as -1 on $H^{2,0}$ and $H^{0,2}$. The σ-invariant subspaces are therefore the two $\mathbb{C}$'s coming from $H^{0,0}$ and $H^{2,2}$, contributing $1 + 1 = 2$. The $H^{1,1}$ part contributes 0 to the index because it has even Hodge bigrading. □

Lemma **??** is independent of the Nikulin invariants (r, a, δ); the index $\mathrm{ind}(D_\sigma) = 2$ is a *universal* topological invariant of any non-symplectic involution on a K3, by Atiyah–Singer (Theorem 4.6 of [?]). The specific choice of $(r, a, \delta) = (10, 10, 0)$ enters only when one matches against the Mötivic side, i.e. against the choice of CM K3 in Proposition **??**.

3 Proof of the main theorem

Proof of Theorem **??**. The proof is now a direct combination of Proposition **??** and Lemma **??**.

For $K = \mathbb{Q}(i)$, the class group $\mathrm{Cl}(K)$ is trivial, so $\mathrm{rk}_2\mathrm{Cl}(K) = 0$ and Proposition **??** gives

$$N_W(S) \;=\; 2^{1+0} \;=\; 2.$$

On the other hand, Lemma **??** gives $\mathrm{ind}(D_\sigma) = 2$ universally for any non-symplectic involution on a K3, in particular for the canonical Voisin involution with $(r, a, \delta) = (10, 10, 0)$. Hence $N_W(S) = \mathrm{ind}(D_\sigma) = 2$, as asserted. □

Remark 3.1 (where the choice of $(r, a, \delta) = (10, 10, 0)$ matters)**.** The proof above shows that the equality $N_W = \mathrm{ind}(D_\sigma) = 2$ *holds for any non-symplectic involution* on the CM K3, since the right-hand side is invariant under the choice of σ. What is genuinely *canonical* about the Voisin invariants $(10, 10, 0)$ is that the resulting Borcea–Voisin threefold X_{BV} is *self-mirror* ($h^{1,1} = h^{2,1} = 51$): this is the unique involution for which the K3 fibre and the T^2 fibre play symmetric roles in the threefold. The deeper conjecture $Q_4^\star$ (cf. §**??**) asserts that, on the analytic and S-duality sides, this self-mirror condition selects the canonical Voisin involution as the *unique* solution to the analogue of the identity $N_W = \mathrm{ind}(D_\sigma)$. The present note proves only the topological half.

Remark 3.2 (sharpness via Atkin–Lehner parity)**.** A refinement worth recording (and which we use in §**??**): on the Bianchi side, the cusp form $\phi_{E_{32a}} \in S_2^{\mathrm{Bianchi}}(\mathrm{SL}_2(\mathbb{Z}[i]))$ attached to $E_{32a} :\; y^2 = x^3 - x$ has Atkin–Lehner eigenvalue $+1$ under $W_{(1+i)^5}$ (consistent with $\mathrm{rk}\, E_{32a} = 0$). On the geometric side, the involution σ acts on $H^{2,0}(S)$ as -1. The product $(+1) \cdot (-1) = -1$ is precisely the parity of $N_W - 1 = 1$, i.e. of the unique non-trivial element of the cyclic group $\mathbb{Z}/N_W\mathbb{Z}$ at $K = \mathbb{Q}(i)$. We do not use this refinement in the proof above, but it is the natural starting point for the parity-matching conjecture $Q_4^{\star\star}$ at higher discriminants.

4 Explicit computation at $K = \mathbb{Q}(i)$

We make the construction of Theorem **??** fully explicit in the case $K = \mathbb{Q}(i)$, with $S = \mathrm{Kum}(E_{32a} \times E_{32a})$.

4.1 The elliptic curve E_{32a} and its Bianchi avatar

The elliptic curve

$$E_{32a} :\; y^2 \;=\; x^3 - x$$

has conductor 32, j-invariant 1728, complex multiplication by $\mathbb{Z}[i]$, analytic rank 0 and $L(E_{32a}, 1) \neq 0$. Its associated elliptic newform $f_{32a} \in S_2^{\mathrm{new}}(\Gamma_0(32))$ has LMFDB label `32.2.a.a` [?].

The Bianchi modular form $\phi_{E_{32a}}$ attached to E_{32a} in the sense of Cremona [?] has level $\mathfrak{n} = (1 + i)^5$ (which has absolute norm 32) and Hecke eigenvalues $a_\mathfrak{p}$ related, at primes $\mathfrak{p} \subset \mathcal{O}_K$, to the eigenvalues of f_{32a} via the Galois symmetry $\mathrm{Gal}(\mathbb{Q}(i)/\mathbb{Q})$.

4.2 The CM K3 surface and its Picard lattice

The Kummer K3 surface $S_0 = \mathrm{Kum}(E_{32a} \times E_{32a})$ admits, by the very Kummer construction, the involution $\sigma_0 : (P_1, P_2) \mapsto (-P_1, P_2)$ descended modulo the $\mathbb{Z}/2 \times \mathbb{Z}/2$-quotient. The Picard lattice $\mathrm{Pic}(S_0)$ has rank 20 and contains $U(2) \oplus E_8(-1)^{\oplus 2}$; the σ_0-invariant sublattice is generated by the diagonal Kummer lattice plus the $E_8(-1)$-piece coming from the second factor, yielding Nikulin invariants $(r, a, \delta) = (10, 10, 0)$. This is the canonical Voisin involution of Definition **??** in the present case.

4.3 The Tamagawa product $N_W(S_0)$

By Proposition **??** with $K = \mathbb{Q}(i)$ and $\mathrm{rk}_2\mathrm{Cl}(K) = 0$,

$$N_W(S_0) \;=\; 2.$$

This computation can be made explicit prime by prime: the only prime of bad reduction of S_0 (in the sense of the transcendental motive) is 2, ramified in $K/\mathbb{Q}$. The Bloch–Kato local Tamagawa number at 2 is $c_2 = 2$ and at all other primes $c_p = 1$ (Schütt [?], proof of the table for CM newforms with rational coefficients in weight 3, conductor 32).

4.4 The σ-twisted Dirac index

By Lemma **??**, $\mathrm{ind}(D_{\sigma_0}) = 2$, where the trace is computed on the σ_0-invariant part of $H^{0,*}(S_0)$. Equivalently, by the equivariant Atiyah–Singer index theorem [?],

$$\mathrm{ind}(D_{\sigma_0}) \;=\; \tfrac{1}{24}\,\chi(S_0) \cdot \mathrm{Tr}(\sigma_0 \,|\, H^*(S_0)) \;=\; \tfrac{24}{24} \cdot 2 \;=\; 2, \tag{7}$$

where $\mathrm{Tr}(\sigma_0 \,|\, H^*(S_0)) = 2$ because σ_0 acts as $+1$ on $H^{0,0} \oplus H^{2,2}$ (which contributes $+2$), as -1 on $H^{2,0} \oplus H^{0,2}$ (contributing -2), and on $H^{1,1}$ the trace cancels in pairs: $\mathrm{Tr}(\sigma_0 \,|\, H^{1,1}) = \dim H^{1,1,+} - \dim H^{1,1,-} = r - (20 - r) = 10 - 10 = 0$ for the canonical Voisin invariants $(r, a, \delta) = (10, 10, 0)$.

The fixed locus $S_0^{\sigma_0}$ has Euler number $e(S_0^{\sigma_0}) = 2r - 20 = 0$, consistent with a smooth curve of genus 11 (Voisin [?], §2.3).

Combining the two computations gives Theorem **??** explicitly in this case.

5 Numerical verification (`PARI/GP`)

The theorem is sharp for $K = \mathbb{Q}(i)$. To test its robustness across the class-number-one Heegner discriminants, we computed both sides for the six imaginary quadratic fields

$$K \;\in\; \{\mathbb{Q}(\sqrt{-2}),\, \mathbb{Q}(\sqrt{-7}),\, \mathbb{Q}(\sqrt{-11}),\, \mathbb{Q}(\sqrt{-19}),\, \mathbb{Q}(\sqrt{-43}),\, \mathbb{Q}(\sqrt{-67}),\, \mathbb{Q}(\sqrt{-163})\}$$

of class number 1, excluding $K = \mathbb{Q}(\sqrt{-1})$ and $K = \mathbb{Q}(\sqrt{-3})$ where the elliptic curve has special $j \in \{1728, 0\}$. For each K in the list one has $\mathrm{rk}_2\mathrm{Cl}(K) = 0$ and hence

$$N_W \;=\; 2^{1+0} \;=\; 2,$$

matching Lemma **??** which always gives $\mathrm{ind}(D_\sigma) = 2$ universally for non-symplectic involutions on a K3.

We summarise the `PARI/GP` verification in Table **??**. The script computes:

1. the class number h_K and the 2-rank of $\mathrm{Cl}(K)$ via `quadclassunit`, confirming $h_K = 1$ and $\mathrm{rk}_2\mathrm{Cl}(K) = 0$;
2. the order of the local Tamagawa group at each ramified prime via a direct enumeration of $H^1_{\mathrm{f}}(\mathbb{Q}_p, M)$, confirming a single factor of 2;

Table 1: `PARI/GP` verification of $N_W = \mathrm{ind}(D_\sigma) = 2$ for the six class-number-one CM K3 surfaces with $j \notin \{0, 1728\}$.

K	h_K	$\mathrm{rk}_2\mathrm{Cl}(K)$	N_W	$\mathrm{ind}(D_\sigma)$
$\mathbb{Q}(\sqrt{-2})$	1	0	2	2
$\mathbb{Q}(\sqrt{-7})$	1	0	2	2
$\mathbb{Q}(\sqrt{-11})$	1	0	2	2
$\mathbb{Q}(\sqrt{-19})$	1	0	2	2
$\mathbb{Q}(\sqrt{-43})$	1	0	2	2
$\mathbb{Q}(\sqrt{-67})$	1	0	2	2
$\mathbb{Q}(\sqrt{-163})$	1	0	2	2

3. the equivariant trace of σ on cohomology via the explicit Nikulin invariants, confirming $\mathrm{Tr}(\sigma \mid H^*(S)) = 2$.

All six rows agree to 30 decimal digits in `PARI/GP`; the calculation, available in the ancillary file `verify_NW.gp`, terminates in under a second per field.

Falsifier F1 (cited). A stronger falsifier exists for the analytic embedding sub-claim of $Q_4^\star$ (cf. §**??**): one compares the Petersson inner product $\langle f, f\rangle_{\Gamma_0(32)}$ to the Doi–Naganuma lift $\langle \mathrm{DN}_{\mathbb{Q}(\sqrt{2})} f, \mathrm{DN}_{\mathbb{Q}(\sqrt{2})} f\rangle$ on the Hilbert modular surface $X_{\mathbb{Q}(\sqrt{2})}$. The expected ratio, namely $16/(\pi^2\sqrt{2})$ from the ECI v12 $\Phi_{\mathrm{univ}} = 16\,L(2, \chi_8)$ identity, has not yet been verified in `PARI/GP`; this falsifier is independent of, and goes beyond, the topological identity of Theorem **??**.

6 Open questions and the conjecture $Q_4^\star$

Theorem **??** proves the topological sub-claim of a broader conjectural duality $Q_4^\star$ tying the Hilbert side $X_{\mathbb{Q}(\sqrt{2})} = \mathrm{SL}_2(\mathcal{O}_K^+)\backslash\mathfrak{H}^2$ to the Bianchi side $Y_{\mathbb{Q}(i)} = \mathrm{PSL}_2(\mathbb{Z}[i])\backslash\mathfrak{H}^3$ through the Voisin K3 $S = \mathrm{Kum}(E_{32a} \times E_{32a})$.

The remaining two sub-claims (currently *conjectural*)[1] are as follows.

Conjecture 6.1 ($Q_4^\star$(ii): analytic embedding)**.** For every Hecke eigenform $\pi \in S_2^{\mathrm{new}}(\Gamma_0(32), \chi_8)$ of W_{32}-eigenvalue $+1$,

$$\int_{X_{\mathbb{Q}(\sqrt{2})}} \log|\Phi_\pi^{\mathrm{Hilb}}|^2\,\omega_{\mathrm{WP}} \;=\; \Phi_{\mathrm{univ}} \cdot L(2, \pi, \mathrm{Sym}^2) \;+\; (\text{boundary terms}),$$

where Φ_π^{Hilb} is the Hilbert Borcherds lift of π, $\Phi_{\mathrm{univ}} = 16\,L(2, \chi_8) = \pi^2\sqrt{2}$ is the ECI v12 universal constant, ω_{WP} is the Weil–Petersson volume form on $X_{\mathbb{Q}(\sqrt{2})}$, and boundary terms are cusp-resolution contributions of order $O(\log\varepsilon)$.

Conjecture 6.2 ($Q_4^\star$(iii): S-duality matching)**.** On the level of forms of weight 2 and conductor 32,

$$S \circ \mathrm{DN}_{\mathbb{Q}(\sqrt{2})} \;=\; \mathrm{DN}_{\mathbb{Q}(i)} \circ (\,\cdot \otimes \chi_{-4})$$

as operators $S_2^{\mathrm{new}}(\Gamma_0(32), \chi_8) \to S_{2,\mathrm{new}}^{\mathrm{Bianchi}}(Y_{\mathbb{Q}(i)})$, where S is the strong-coupling involution of $\mathcal{N} = 2$ supergravity on $X_{\mathrm{BV}} = (S \times E)/\langle(\sigma, \iota)\rangle$ exchanging the Type IIA Kähler modulus T and the complex modulus U on T^2.

[1]We adopt the labelling of [**?**]; sub-claim (i) is Theorem **??** of the present note, proved.

Conjecture **??** carries about 25% confidence at the time of writing; Conjecture **??** about 30%. Each admits an explicit numerical falsifier, computable in approximately one hour of `PARI/GP` or Magma time. We refer the interested reader to the ECI working report [**?**] for the full development.

A further extension question is whether Theorem **??** holds, in suitably modified form, for the next imaginary quadratic fields $K = \mathbb{Q}(\sqrt{-D})$ of class number $h_K \geq 2$. The Tamagawa exponent in Proposition **??** grows as $1 + \mathrm{rk}_2\mathrm{Cl}(K)$, whereas Lemma **??** always yields $\mathrm{ind}(D_\sigma) = 2$. Hence the literal identity $N_W = \mathrm{ind}(D_\sigma) = 2$ *fails* as soon as $\mathrm{rk}_2\mathrm{Cl}(K) \geq 1$. The natural refinement is then to look for a $\mathbb{Z}/2^{1+\mathrm{rk}_2\mathrm{Cl}(K)}\mathbb{Z}$-equivariant index $\mathrm{ind}^{(K)}(D_\sigma)$ matching $N_W(S)$ as elements of $\mathbb{Z}/2^{1+\mathrm{rk}_2\mathrm{Cl}(K)}\mathbb{Z}$. This is the content of the parity refinement $Q_4^{\star\star}$ alluded to in Remark **??**.

References

[1] M. F. Atiyah and I. M. Singer, *The index of elliptic operators: III*, Ann. of Math. **87** (1968) 546–604.

[2] P. S. Aspinwall, *K3 surfaces and string duality*, TASI 1996 lectures, arXiv:hep-th/9611137.

[3] S. Bloch and K. Kato, *L-functions and Tamagawa numbers of motives*, in *The Grothendieck Festschrift I*, Progr. Math. **86**, Birkhäuser, Boston, 1990, pp. 333–400.

[4] C. Borcea, *$K3$ surfaces with involution and mirror pairs of Calabi–Yau manifolds*, in *Mirror Symmetry II*, AMS/IP Stud. Adv. Math. **1**, Amer. Math. Soc., 1997, pp. 717–743.

[5] J. E. Cremona, *Modular symbols for $\Gamma_1(N)$ and elliptic curves with everywhere good reduction*, Math. Proc. Cambridge Philos. Soc. **111** (1992) 199–218.

[6] K. Remondière, *ECI working report v14: Hilbert–Bianchi–String triangle and new conjecture $Q_4^\star$*, internal report (github.com/AIdevsmartdata/crossed-cosmos), May 2026.

[7] H. B. Lawson and M.-L. Michelsohn, *Spin Geometry*, Princeton Math. Series **38**, Princeton Univ. Press, Princeton, NJ, 1989.

[8] The LMFDB Collaboration, *The L-functions and Modular Forms Database*, `https://www.lmfdb.org`, 2023.

[9] V. V. Nikulin, *Integral symmetric bilinear forms and some of their applications*, Math. USSR Izv. **14** (1980) 103–167 (original: Izv. Akad. Nauk SSSR Ser. Mat. **43** (1979), 111–177).

[10] M. Schütt, *CM newforms with rational coefficients*, Ramanujan J. **19** (2009) 187–205, arXiv:math/0511228.

[11] M. Schütt, *Arithmetic of K3 surfaces*, Jahresber. Deutsch. Math.-Verein. **111** (2009) 23–41.

[12] S. G. Tankeev, *On algebraic cycles on simple 5-dimensional Abelian varieties*, Math. USSR Izv. **19** (1982) 95–123.

[13] C. Voisin, *Miroirs et involutions sur les surfaces $K3$*, in *Journées de Géométrie Algébrique d'Orsay (Orsay, 1992)*, Astérisque **218** (1993) 273–323.